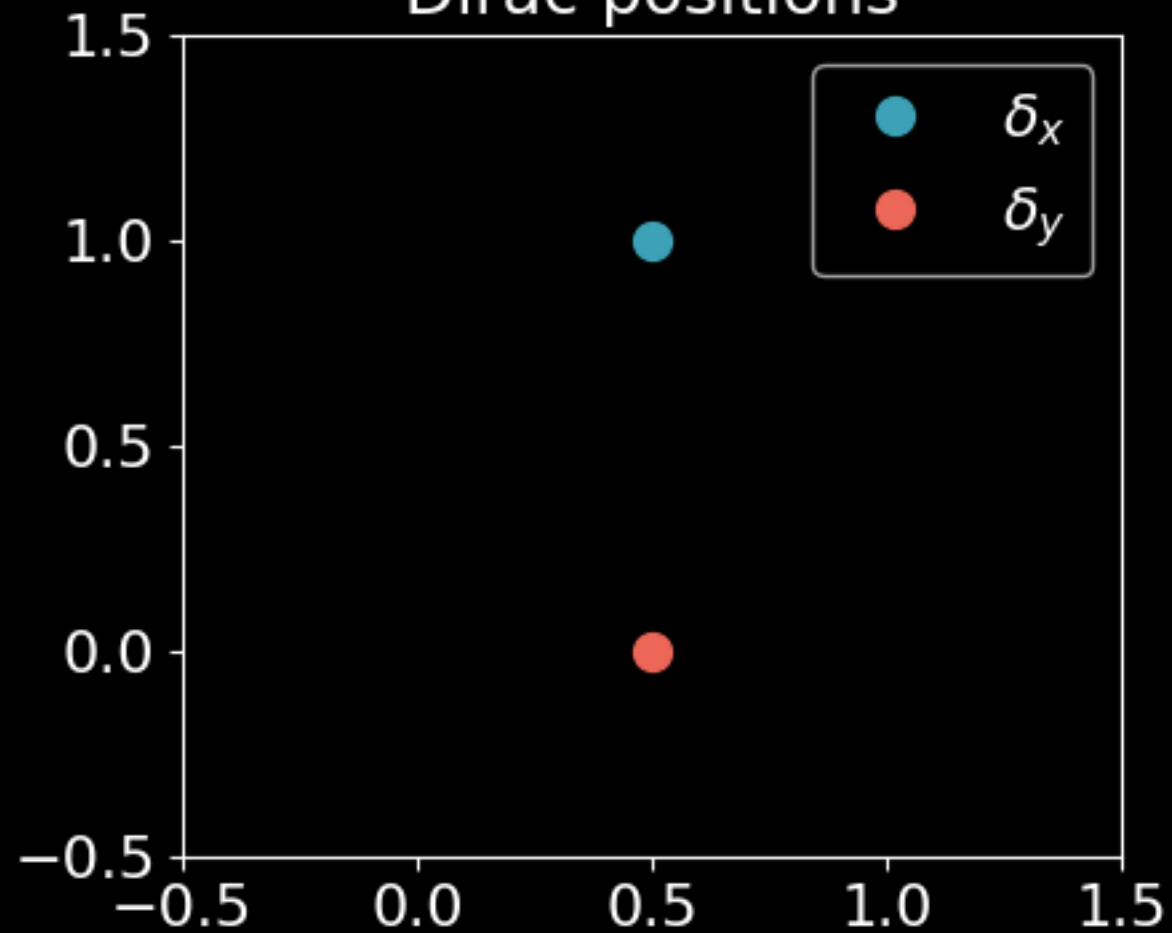
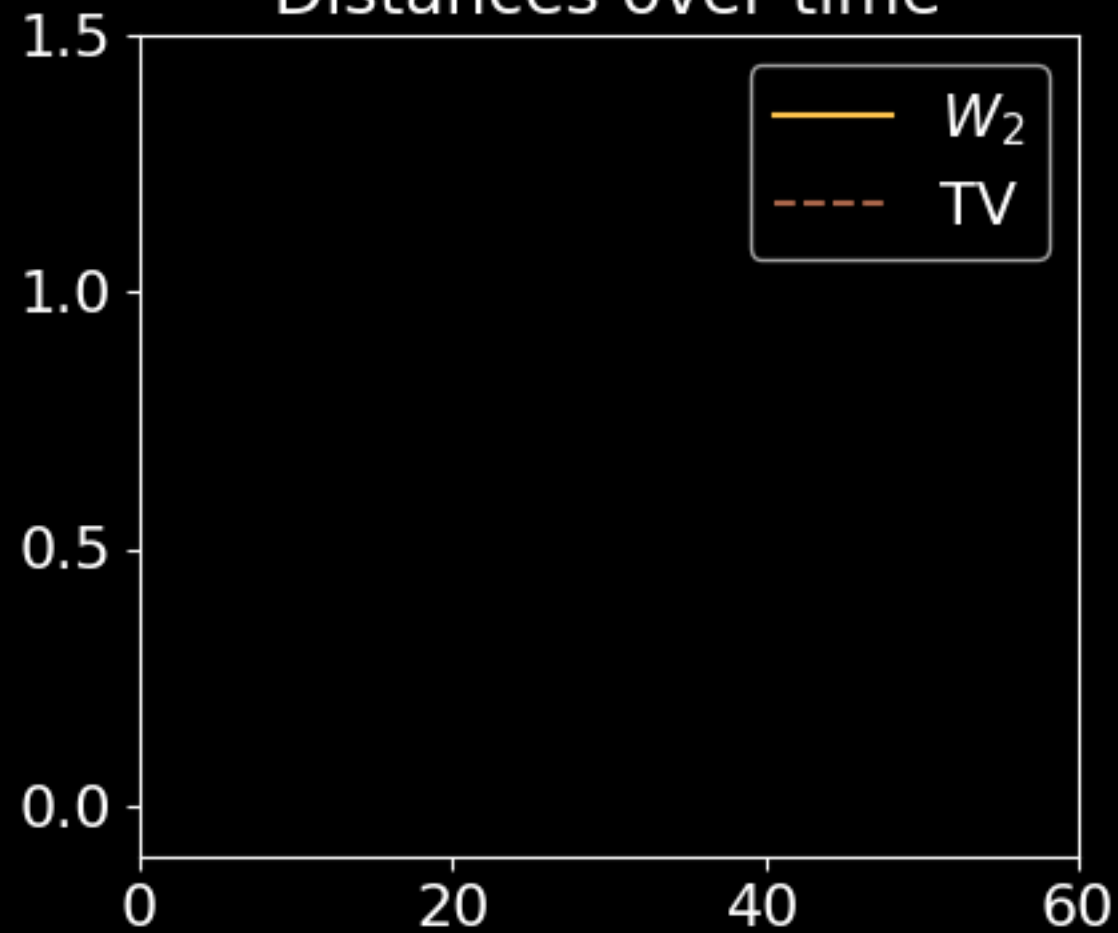


Dirac positions



Distances over time

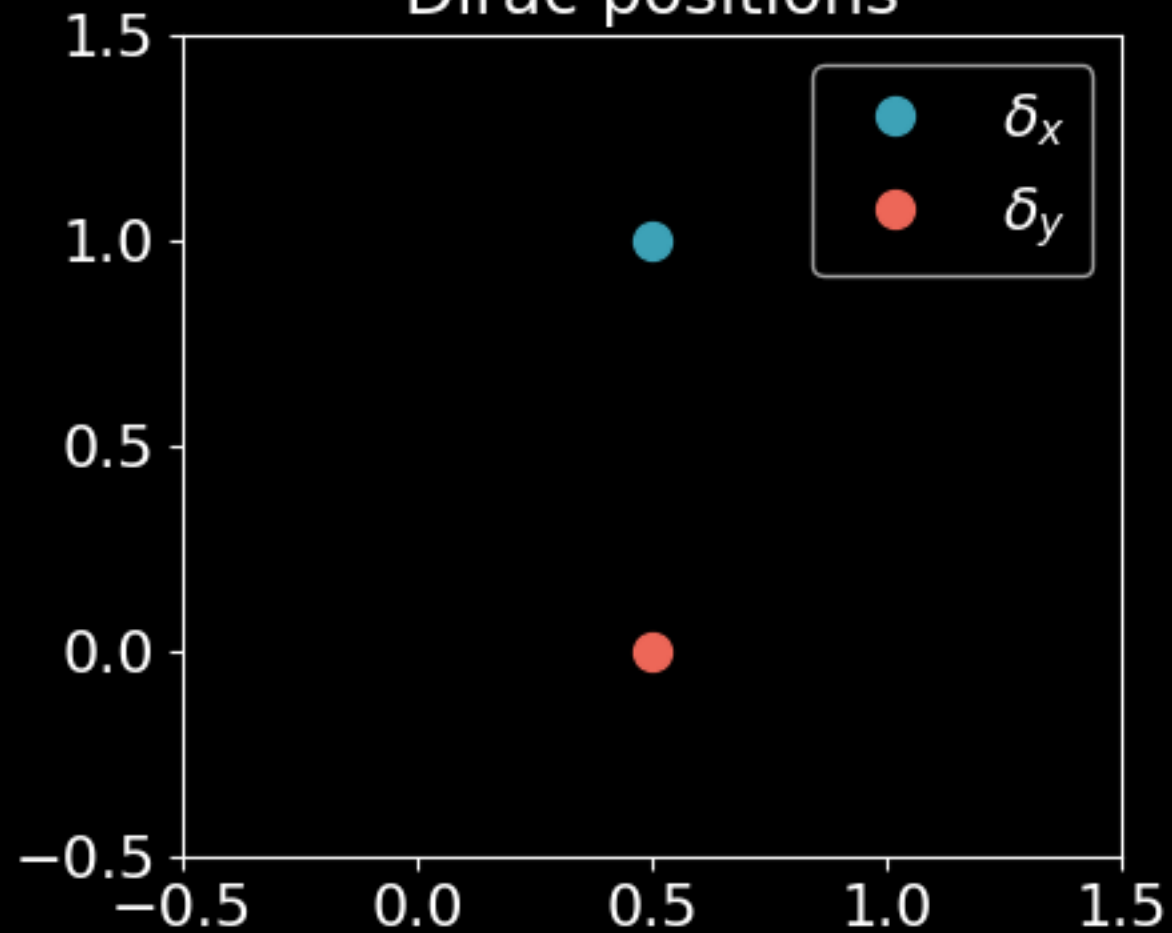




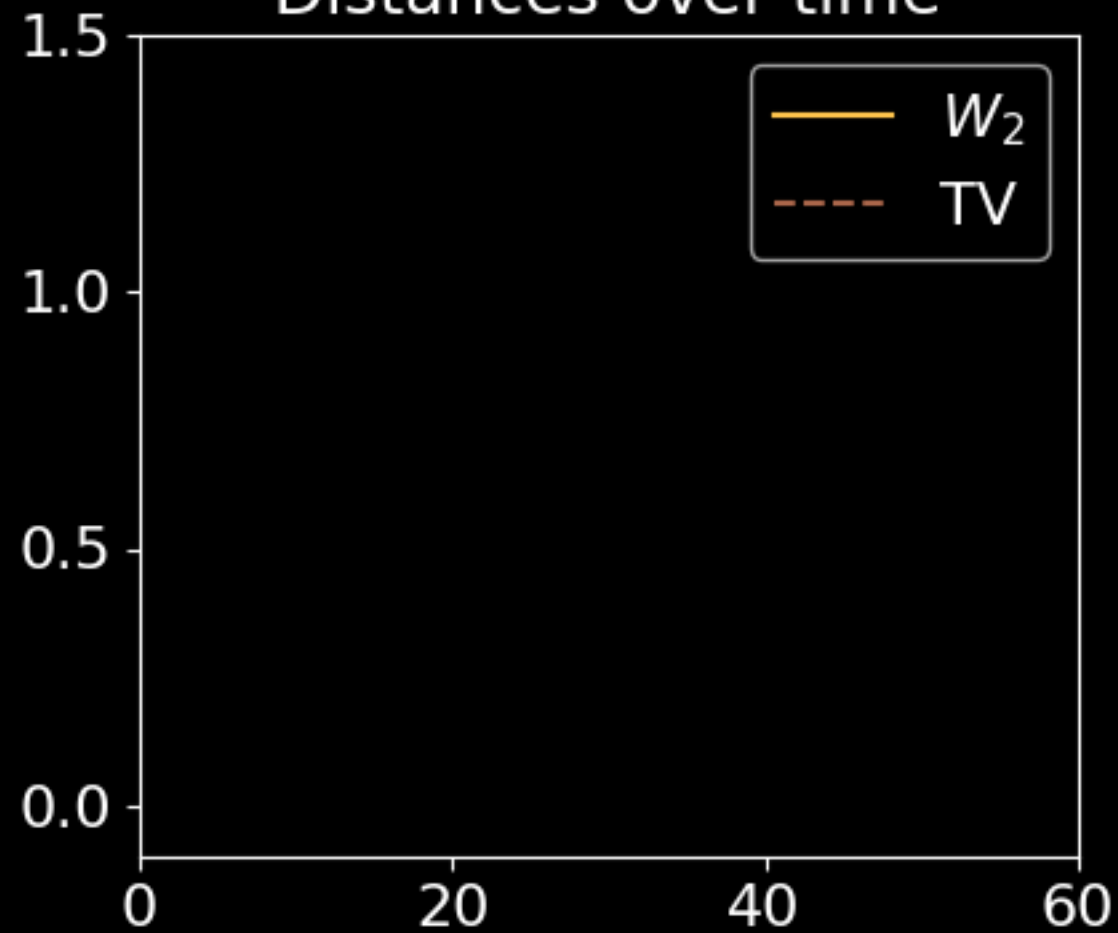




Dirac positions



Distances over time

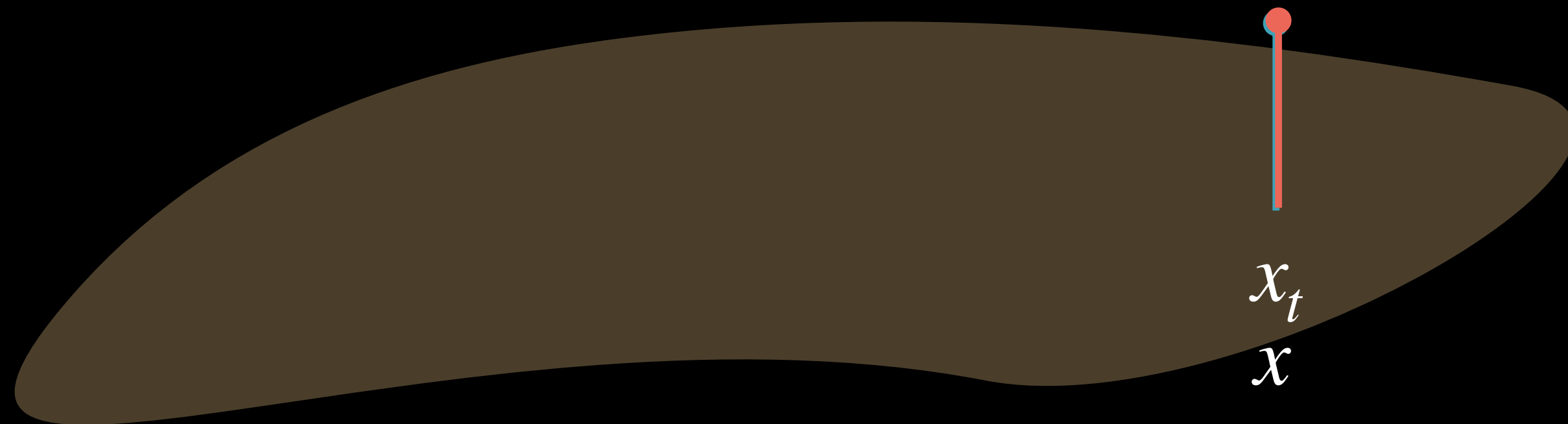


Why geometry awareness matters

$$\text{TV}(\delta_{x_t}, \delta_x) = 1$$

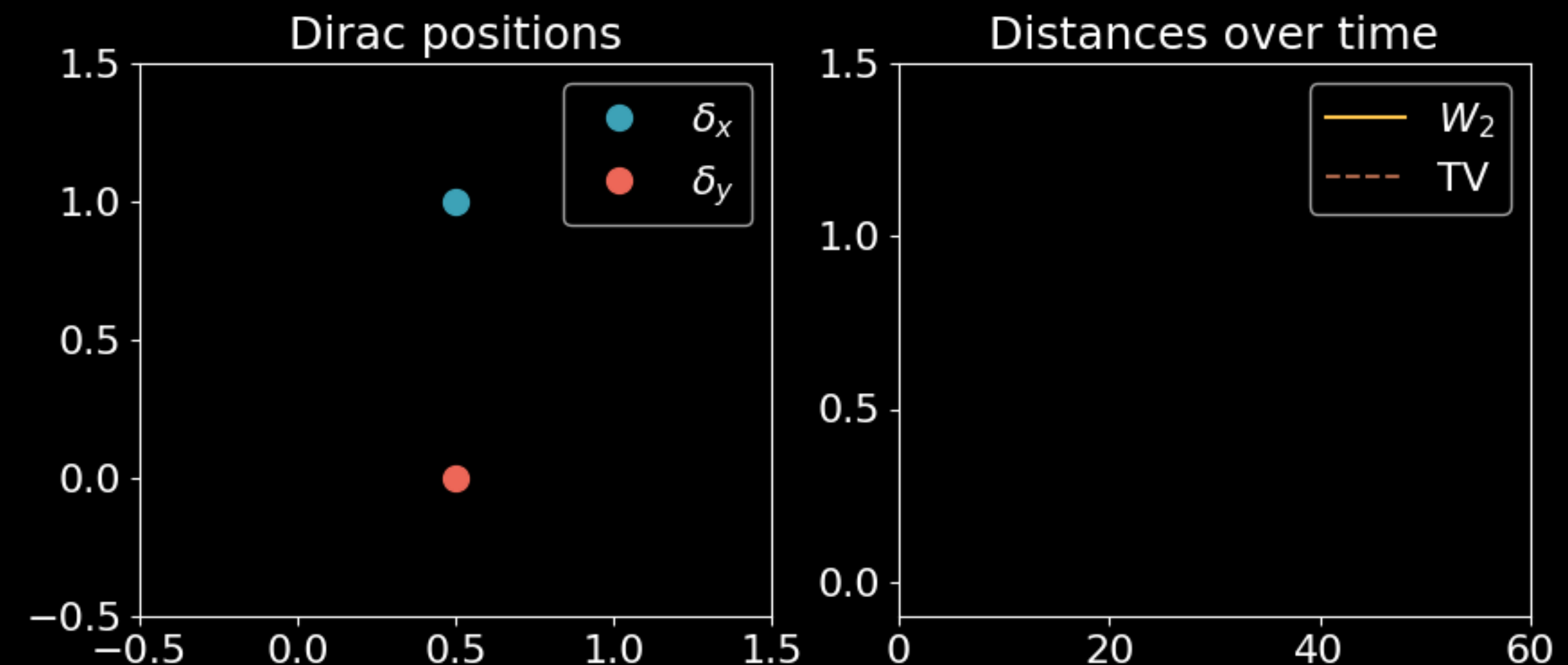
$$\text{KL}(\delta_{x_t}, \delta_x) = \infty$$

$$W_p(\delta_{x_t}, \delta_x) = d(x_t, x)$$



More generally:

- KL undefined if supports don't overlap
- TV don't consider geometry
- OT/Wasserstein respects underlying space!



Geometry: KL vs OT